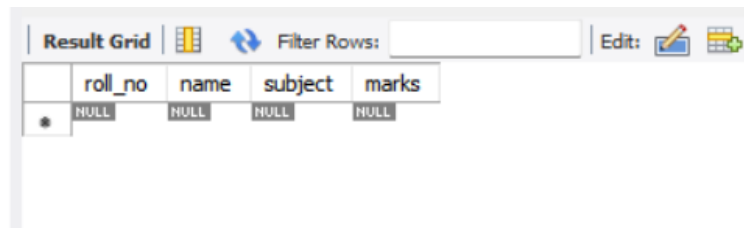


Task 2 Outputs

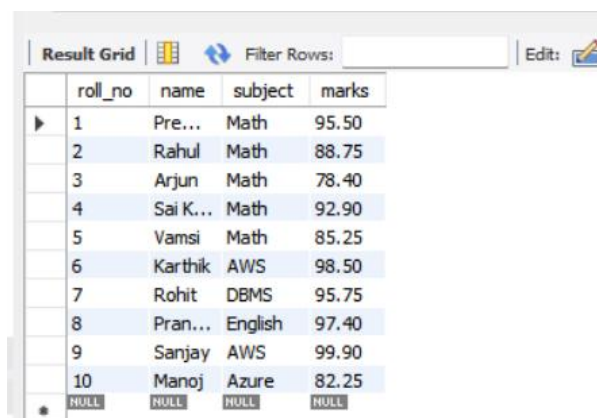
```
CREATE TABLE student_marks (  
    roll_no INT PRIMARY KEY,  
    name VARCHAR(50),  
    subject VARCHAR(50),  
    marks DECIMAL(5,2)  
);
```



The screenshot shows a database interface with a 'Result Grid' tab. The table has four columns: roll_no, name, subject, and marks. The first row contains NULL values for all columns. The interface includes a 'Filter Rows' search bar and an 'Edit' button.

	roll_no	name	subject	marks
*	NULL	NULL	NULL	NULL

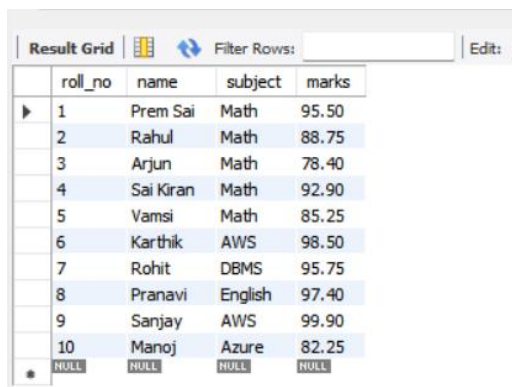
```
INSERT INTO student_marks (roll_no, name, subject, marks) VALUES  
(1, 'Prem Sai', 'Math', 95.50),  
(2, 'Rahul', 'Math', 88.75),  
(3, 'Arjun', 'Math', 78.40),  
(4, 'Sai Kiran', 'Math', 92.90),  
(5, 'Vamsi', 'Math', 85.25),  
(6, 'Karthik', 'AWS', 98.50),  
(7, 'Rohit', 'DBMS', 95.75),  
(8, 'Pranavi', 'English', 97.40),  
(9, 'Sanjay', 'AWS', 99.90),  
(10, 'Manoj', 'Azure', 82.25);
```



The screenshot shows the same database interface as before, but now with 10 rows of data. The columns are roll_no, name, subject, and marks. The data is as follows:

	roll_no	name	subject	marks
▶	1	Pre...	Math	95.50
	2	Rahul	Math	88.75
	3	Arjun	Math	78.40
	4	Sai K...	Math	92.90
	5	Vamsi	Math	85.25
	6	Karthik	AWS	98.50
	7	Rohit	DBMS	95.75
	8	Pran...	English	97.40
	9	Sanjay	AWS	99.90
	10	Manoj	Azure	82.25
*	NULL	NULL	NULL	NULL

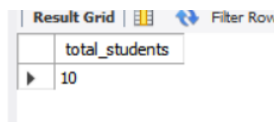
```
SELECT * FROM student_marks;
```



	roll_no	name	subject	marks
▶	1	Prem Sai	Math	95.50
	2	Rahul	Math	88.75
	3	Arjun	Math	78.40
	4	Sai Kiran	Math	92.90
	5	Vamsi	Math	85.25
	6	Karthik	AWS	98.50
	7	Rohit	DBMS	95.75
	8	Pranavi	English	97.40
	9	Sanjay	AWS	99.90
	10	Manoj	Azure	82.25
•	NULL	NULL	NULL	NULL

```
SELECT COUNT(*) AS total_students
```

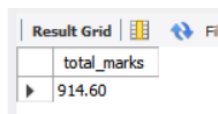
```
FROM student_marks;
```



	total_students
▶	10

```
SELECT SUM(marks) AS total_marks
```

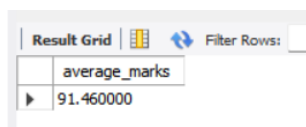
```
FROM student_marks;
```



	total_marks
▶	914.60

```
SELECT AVG(marks) AS average_marks
```

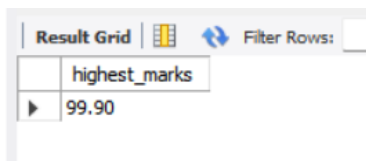
```
FROM student_marks;
```



	average_marks
▶	91.460000

```
SELECT MAX(marks) AS highest_marks
```

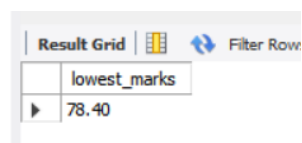
```
FROM student_marks;
```



	highest_marks
▶	99.90

```
SELECT MIN(marks) AS lowest_marks
```

```
FROM student_marks;
```



	lowest_marks
▶	78.40

```
SELECT *
FROM student_marks
WHERE marks > 85;
```

Result Grid  Filter Rows:				
	roll_no	name	subject	marks
▶	1	Prem Sai	Math	95.50
	2	Rahul	Math	88.75
	4	Sai Kiran	Math	92.90
	5	Vamsi	Math	85.25
	6	Karthik	AWS	98.50
	7	Rohit	DBMS	95.75
	8	Pranavi	English	97.40
	9	Sanjay	AWS	99.90
✱	NULL	NULL	NULL	NULL

```
SELECT *
FROM student_marks
WHERE marks >= 90;
```

Result Grid  Filter Rows:				
	roll_no	name	subject	marks
▶	1	Prem Sai	Math	95.50
	4	Sai Kiran	Math	92.90
	6	Karthik	AWS	98.50
	7	Rohit	DBMS	95.75
	8	Pranavi	English	97.40
	9	Sanjay	AWS	99.90
✱	NULL	NULL	NULL	NULL

```
SELECT *
FROM student_marks
WHERE marks < 80;
```

Result Grid  Filter Rows: Edit:				
	roll_no	name	subject	marks
▶	3	Arjun	Math	78.40
✱	NULL	NULL	NULL	NULL

```
SELECT *
FROM student_marks
WHERE marks BETWEEN 80 AND 90;
```

Result Grid  Filter Rows:				
	roll_no	name	subject	marks
▶	2	Rahul	Math	88.75
	5	Vamsi	Math	85.25
	10	Manoj	Azure	82.25
✱	NULL	NULL	NULL	NULL

```
SELECT *
FROM student_marks
WHERE name LIKE 'P%';
```

Result Grid				
	roll_no	name	subject	marks
▶	1	Prem Sai	Math	95.50
	8	Pranavi	English	97.40
✱	NULL	NULL	NULL	97.40

```
SELECT *
FROM student_marks
WHERE name IN ('Prem Sai', 'Rahul', 'Vamsi');
```

Result Grid				
	roll_no	name	subject	marks
▶	1	Prem Sai	Math	95.50
	2	Rahul	Math	88.75
	5	Vamsi	Math	85.25
✱	NULL	NULL	NULL	NULL

```
SELECT *
FROM student_marks
WHERE marks > 85
AND (subject = 'Math' OR name LIKE 'P%');
```

Result Grid				
	roll_no	name	subject	marks
▶	1	Prem Sai	Math	95.50
	2	Rahul	Math	88.75
	4	Sai Kiran	Math	92.90
	5	Vamsi	Math	85.25
	8	Pranavi	English	97.40
✱	NULL	NULL	NULL	NULL

```
UPDATE student_marks
SET marks = 90.00
WHERE roll_no = 3;
```

	3	Arjun	Math	90.00
--	---	-------	------	-------

```
UPDATE student_marks
SET subject = 'Remedial Math'
WHERE marks < 80;
```

4	Sai K...	Math	92.90
6	Karthik	AWS	98.50

DELETE FROM student_marks

WHERE roll_no = 5;

DELETE FROM student_marks

WHERE marks < 75;

SELECT *

FROM student_marks

ORDER BY marks ASC;

Result Grid				
Filter Rows:				
	roll_no	name	subject	marks
▶	3	Arjun	Math	90.00
	6	Karthik	AWS	98.50
	10	Manoj	Azure	82.25
	8	Pranavi	English	97.40
	1	Prem Sai	Math	95.50
	2	Rahul	Math	88.75
	7	Rohit	DBMS	95.75
	4	Sai Kiran	Math	92.90
	9	Sanjay	AWS	99.90
•	NULL	NULL	NULL	NULL

SELECT *

FROM student_marks

ORDER BY marks DESC;

Result Grid				
Filter Rows:				
Edit:				
	roll_no	name	subject	marks
▶	9	Sanjay	AWS	99.90
	4	Sai Kiran	Math	92.90
	7	Rohit	DBMS	95.75
	2	Rahul	Math	88.75
	1	Prem Sai	Math	95.50
	8	Pranavi	English	97.40
	10	Manoj	Azure	82.25
	6	Karthik	AWS	98.50
	3	Arjun	Math	90.00
•	NULL	NULL	NULL	NULL

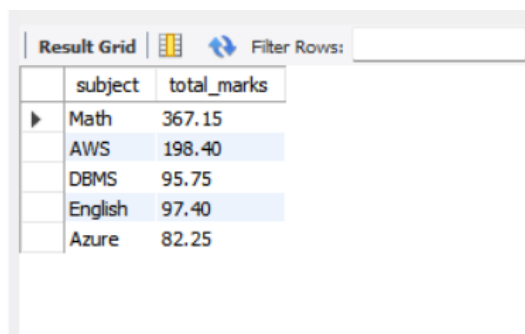
SELECT *

FROM student_marks

ORDER BY name ASC;

```
SELECT *  
FROM student_marks  
ORDER BY name DESC;
```

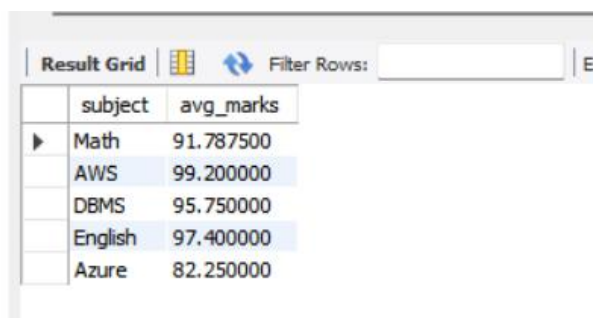
```
SELECT subject,  
       SUM(marks) AS total_marks  
FROM student_marks  
GROUP BY subject;
```



The screenshot shows a 'Result Grid' window with a 'Filter Rows' input field. The grid contains the following data:

	subject	total_marks
▶	Math	367.15
	AWS	198.40
	DBMS	95.75
	English	97.40
	Azure	82.25

```
SELECT subject,  
       AVG(marks) AS avg_marks  
FROM student_marks  
GROUP BY subject;
```



The screenshot shows a 'Result Grid' window with a 'Filter Rows' input field. The grid contains the following data:

	subject	avg_marks
▶	Math	91.787500
	AWS	99.200000
	DBMS	95.750000
	English	97.400000
	Azure	82.250000

```
SELECT subject,  
       COUNT(*) AS num_students  
FROM student_marks  
GROUP BY subject;
```

Result Grid			Filter Rows:
	subject	num_students	
▶	Math	4	
	AWS	2	
	DBMS	1	
	English	1	
	Azure	1	

```

SELECT subject,
       AVG(marks) AS avg_marks
FROM student_marks
GROUP BY subject
HAVING AVG(marks) > 90;

```

Result Grid			Filter Rows:	Export:
	subject	avg_marks		
▶	Math	91.787500		
	AWS	99.200000		
	DBMS	95.750000		
	English	97.400000		

```

SELECT subject,
       COUNT(*) AS num_students
FROM student_marks
GROUP BY subject
HAVING COUNT(*) > 1;

```

Result Grid			Filter Rows:	Export:	Wrap Cell C
	subject	num_students			
▶	Math	4			
	AWS	2			